Using Google Colab and GitHub

Spring 2026

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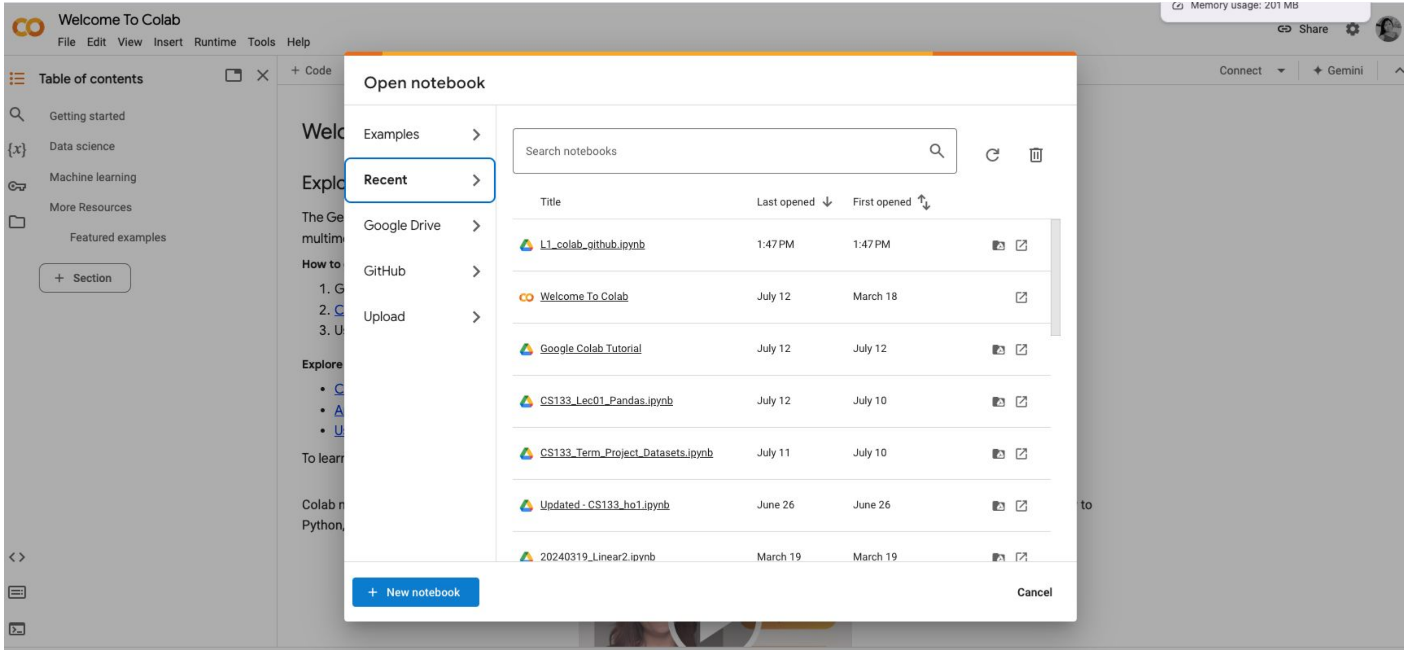
SJSU CS 133

Why are we using Google Colab?

- Cloud-based Jupyter notebook environment
- No setup, easy sharing/collaborating
- Write and execute Python code
- Free GPU and TPU (eg., TensorFlow, PyTorch)
- Import datasets from Kaggle
- Import to GitHub
- Notebooks are ideal for data analysis because they can include:
  - Nicely formatted text (e.g., text that describes the data)
  - Code and code output
  - Spaces for displaying plots

Getting started

- Need Google account, GitHub account and Kaggle (optional)
- Open https://colab.research.google.com and click on + New notebook



Notebook example

- Pull the GitHub repository to make a copy of the in-class hands-on notebook.

- Each bit of text or code is referred to as a **cell**

Note that code cells can also contain **comments**, which are lines of text that are *not* evaluated as code. Comments are used to write brief notes about specific lines of code, and are preceded by the `#` symbol. In the example below I've included a comment and some code.

Exercise 14

Run the cell to see what happens.

Text cell

```
[ ]
# This is a comment: 10 - 5
10 - 2
```

Code cell

Example: Sample notebook

Each bit of text or code is referred to as a **cell**

Sample notebook

This is text! You can make things **bold** or *italic*, you can include links, and you can make bulleted lists:

- The is the first bullet
- This is the second
- Last one

Text cell

[1]
```
2 + 2
```
Code

```
4
```
Output

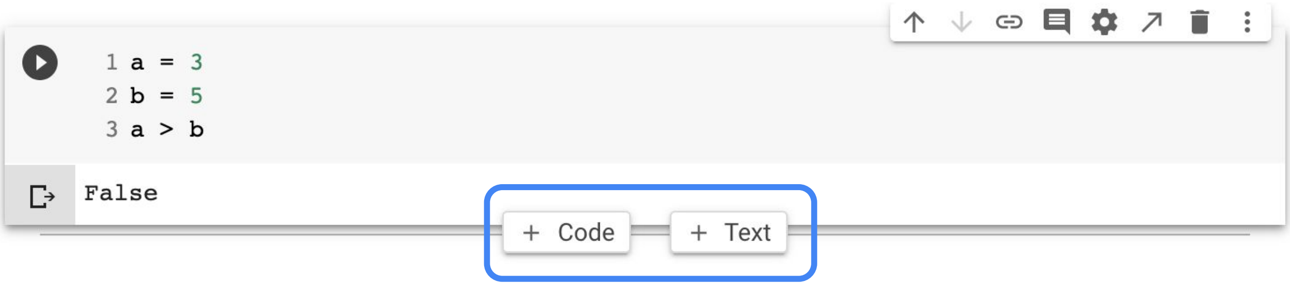
[3]
```
a = 3
b = 5
a > b
```
Code

```
False
```
Output

Code cell

Creating new cells

- Some exercises will ask you to create a new cell
  - Single click on an existing text/code cell, and then hover your mouse towards the bottom edge of the cell, in the center
  - You can select either a Code cell or Text cell

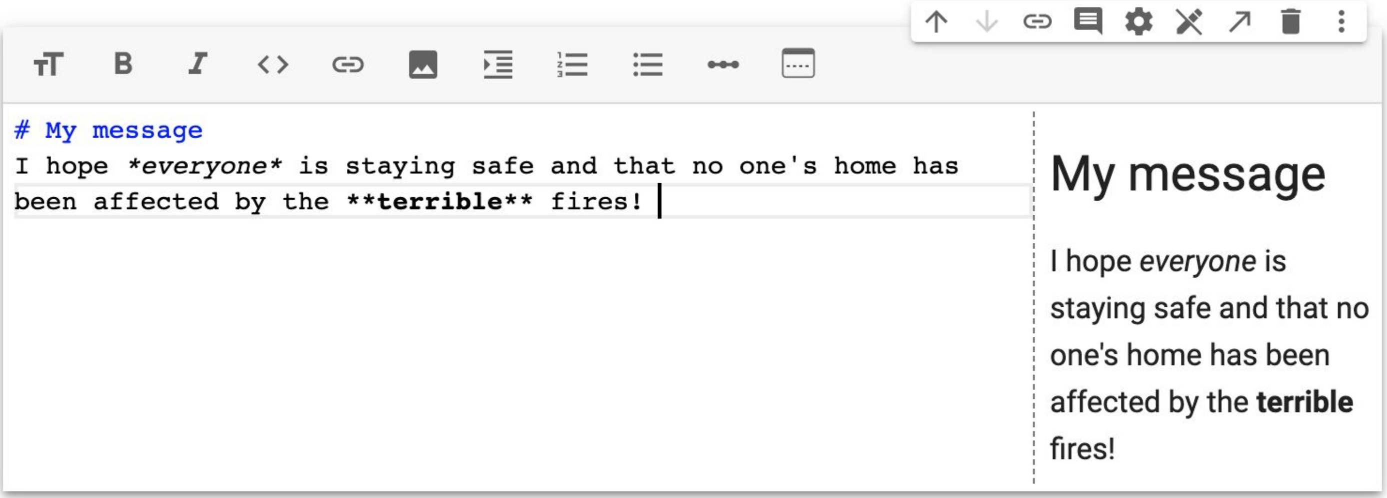


Formatting text cells

- Colab has buttons for things like:
  - Changing font size
  - Making things **bold** and *italic*
  - Creating numbered/bulleted lists
  - Creating links

- Note: you can make these formatting changes by typing, i.e., without clicking buttons
  - As you work through the hands-on, you'll learn how to do this

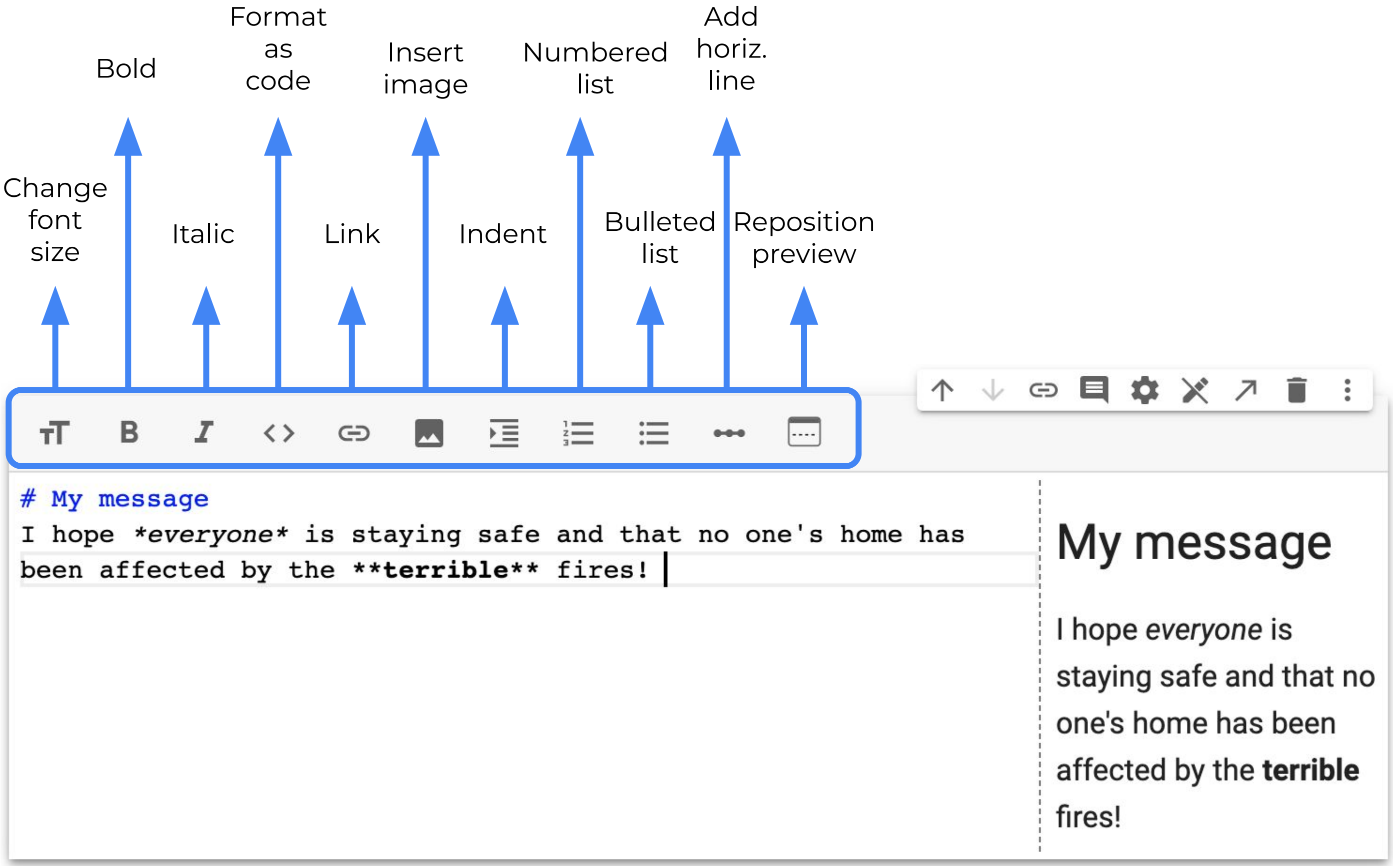
Formatting text cells



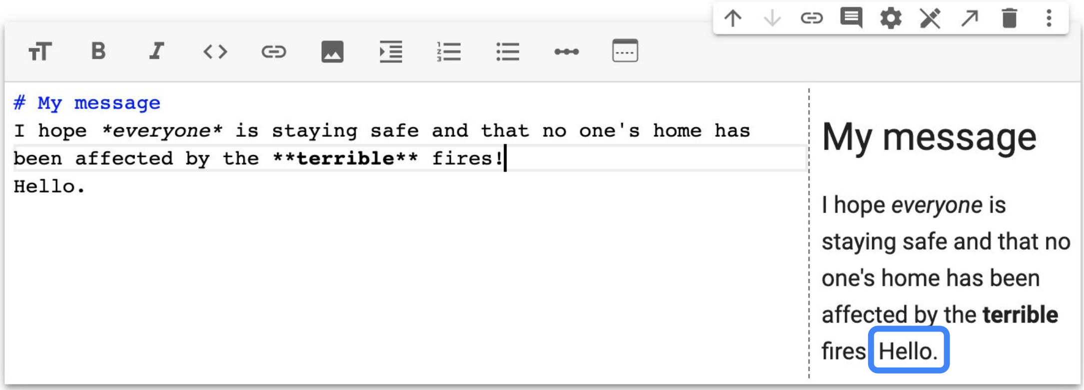
You'll type here,
on the left

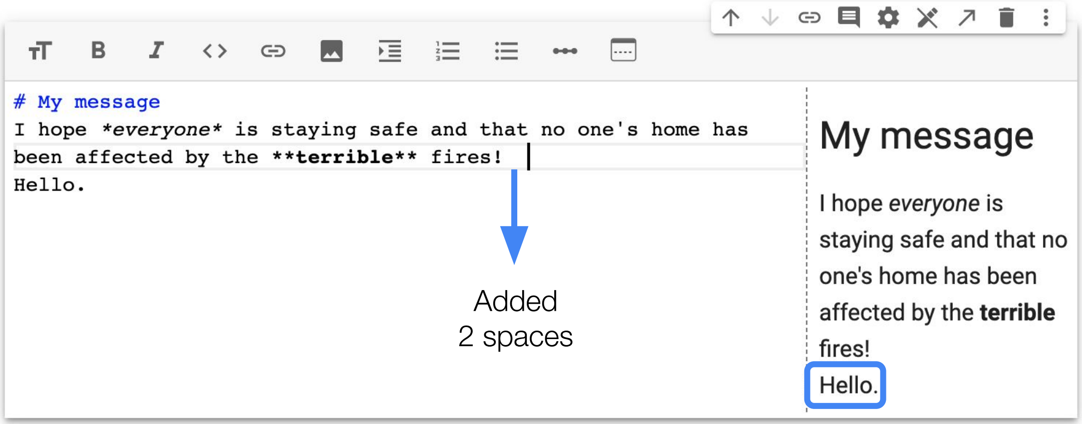
On the right
is a preview

Formatting text cells



Forcing new lines in text cells





To create a new line inside a text cell, use 2 spaces ␣ ␣ at the end of the line, then hit enter

“Running” a cell

When editing a text cell, the left side looks rather ugly. To make things look nice (like the preview), we need to “run” the cell.

Simultaneously hit **`Shift`** and **`Enter`** to run a cell

-or-

Hit the (play) button

This is true for code cells too.

Code cells to run python scripts

- Code cells can run simple and complex Python code
- An example is using Python as a calculator to run simple mathematical expressions
  - Type in a simple mathematical expression and then run the cell
  - The output (found under the code cell) will be the answer

```
[1]  2 + 2
```

```
4
```

```
[3]  a = 3
     b = 5
     a > b
```

```
False
```

Comments within code cells

- Within a code cell, you can also include comments about the code
- Comments: Lines of text within a code cell that are *not* evaluated as code
  - Used for writing brief notes about certain lines of code
  - Preceded by the # symbol
  - For multi-line comments use ‘’’ ‘’’

Comments within code cells

- When running the following cell, we see the output of the code (line 2) and not the comment (line 1)

```
[5]   1 # 5 + 1
      2 2.6 * 3.4
```

```
8.84
```

- Note: Please put a space after the # sign when writing comments

Text vs. Comments

When should you use a text cell, and when should you write a comment in a code cell?

- Comments
  - Generally very short (less than a sentence)
  - Used to provide notes about specific lines of code
- Text cells
  - Appropriate for longer text or text that is formatted

Text vs. Comments

Practice notebook

In this notebook, I'll be practicing some simple Python commands. For example:

- Conditional execution
- Iteration

You'll be learning both of these skills later in the semester!

```
# conditional execution
x = 3
if x > 1:
    print('Greater than 1')
else:
    print('Less than or equal to 1')

# iteration
my_list = [1, 2, 3]
for num in my_list:
    print(num)
```

```
Greater than 1
1
2
3
```

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Text vs. Comments

Notebook documentation

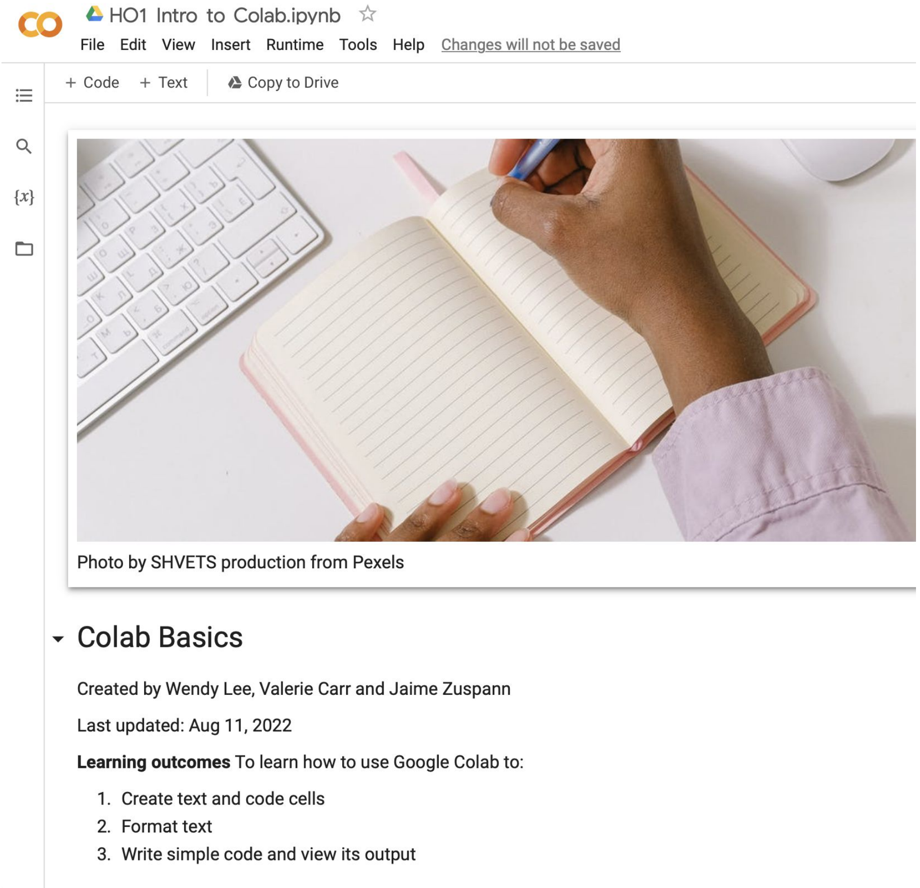
HO1 Intro to Colab.ipynb
File Edit View Insert Runtime Tools Help Changes will not be saved
+ Code + Text Copy to Drive

Photo by SHVETS production from Pexels

Colab Basics

Created by Wendy Lee, Valerie Carr and Jaime Zuspann

Last updated: Aug 11, 2022

Learning outcomes To learn how to use Google Colab to:

1. Create text and code cells
2. Format text
3. Write simple code and view its output

At the top of every notebook, it's helpful to include details like:

- Notebook title, using heading 1
- Name of person who created the notebook (or your name and SJSU ID, if it's an assignment)
- Date that you last updated the notebook
- Purpose of the notebook